Introduction to Health Economics

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- What is moral hazard?
  - Moral hazard is the tendency for insurance against loss to reduce incentives to prevent or minimize the cost of loss (Baker 1996).
  - Different from natural hazards (like earthquake) that cause losses.
  - Moral hazards occur due to carelessness and fraud that can lead to losses and are the result of decisions by humans.
- Story of A, B and C about their experience of paragliding.

- Each case of moral hazard follows a simple pattern:
  - An individual faces some risk of a bad event X, and his actions can increase or decrease its likelihood.
  - The individual buys an insurance contract that will help pay some or all of the costs of X if it occurs. The price of X to the individual is now lower – this price distortion means the individual does not face all the consequences of his actions.
  - In response to the price distortion, the individual changes his behavior in a way that increases the chances of X or increases the costs of recovering from X.
  - The insurance company cannot observe this behavior change, because there is an information asymmetry. Otherwise the contract would have been written to discourage or penalize the riskier behavior.
  - The individual's riskier behavior creates a social loss, because the costly event X occurs more than it would have without insurance.
- Moral hazard is the downside of health insurance because it raises society's level of health care expenditures.

- There are two types of moral hazard:
  - Ex ante moral hazard: behavior changes that occur before an insured event happens and make that event more likely. Examples: going for paragliding.
  - Ex post moral hazard: behavior changes that occur after an insured event happens and make recovering from that event more expensive. Examples taking an expensive drug like miracle pill for fast recovery rather than a more inexpensive remedy.
- How does moral hazard lead to social loss?

- Consider an individual who takes insurance against the medication of diabetes.
  - Without health insurance: her actual cost of medicine is P_U
  - With health insurance: the cost of medicine faced by her declines to P_I.
  - Her dietary plan helps in determining the sugar level in the blood. Insurance cover might induce her to take less precautions to control her sugar level as part of the cost of medicines is borne by the insurance company.

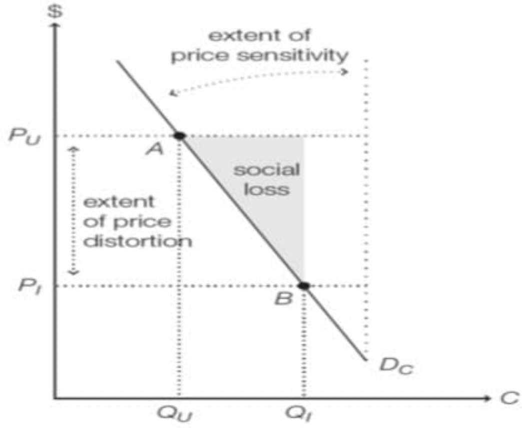

Figure 1: Analysis of social loss caused by moral hazard

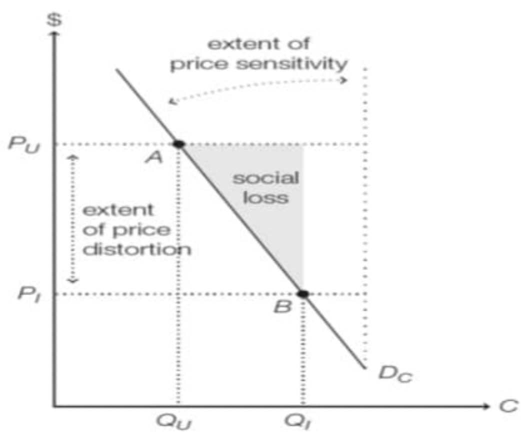

Figure 2: Analysis of social loss caused by moral hazard

- Point A is the socially efficient equilibrium, while Point B is the outcome with insurance.

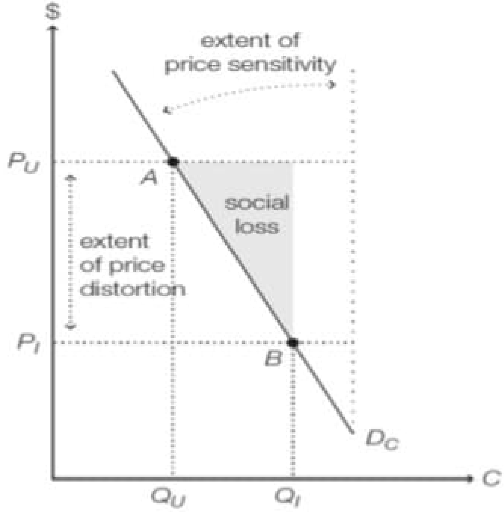

Figure 3: Analysis of social loss caused by moral hazard

- The vertical distance between P_U and P_I shows the extent of price distortion.
  - Completeness of the insurance determines the price distortion.
- The angle between the demand curve D_C and the vertical line represents the extent of price sensitivity.
  - The extent of price sensitivity depends mostly on the nature of the risk being insured, and how controllable it is. For example - health care against genetic disease

- Varying price distortion and price sensitivity

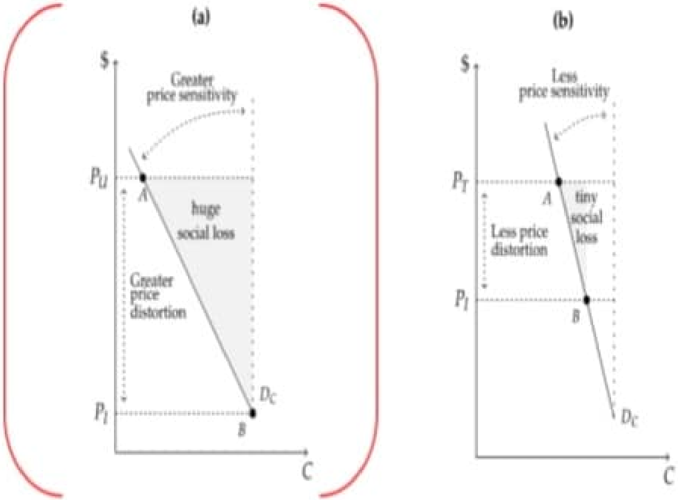

Figure 4: Varying the extent of price distortion and price sensitivity affects the social loss from moral hazard.

- In Figure (a), price distortion and price sensitivity are both quite high. Insured individuals bear little of the cost of their medicine to control blood sugar level, and as they are quite price-sensitive they respond with less care and precaution leading to extra need of medicine. This results in a large social loss.

- Varying price distortion and price sensitivity

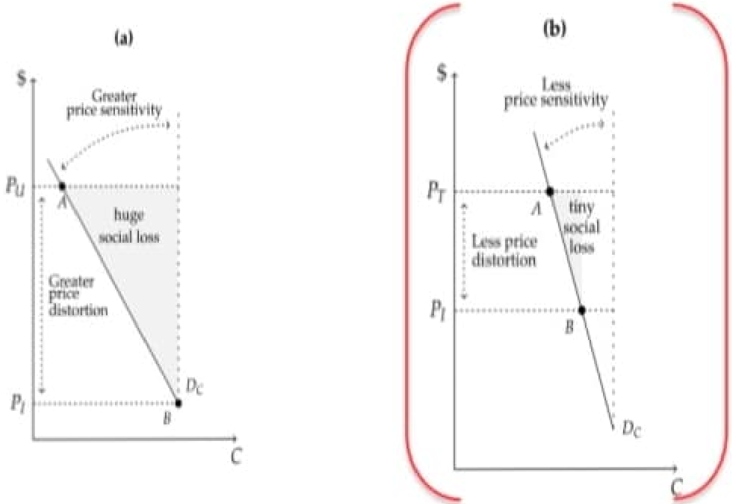

Figure 5: Varying the extent of price distortion and price sensitivity affects the social loss from moral hazard.

- In Figure (b), price distortion and price sensitivity are minimal. Insured individuals bear most of the cost of the medicine of skin infection, and their demand for medicine of skin infection is not very sensitive to prices anyway. In this case, there is still moral hazard but it produces a much smaller social loss.

- Insurance against different types of risks tends to create different amounts of moral hazard.
  - An insurance policy against developing a rare skin disease determined entirely by one's genetic makeup will not induce moral hazard because it does not affect the insured party's actions or decisions which may affect the severity of this disease.
  - An insurance policy against developing a heart attack may induce moral hazard because as an individual would prefer consuming junk food if she knows that the part of cost of health care in case of heart attack will be borne by the insurance company. Over consumption of junk food might increase the likelihood of heart attack leading to moral hazard.
  - If there is either no price distortion or no behavioral changes, then moral hazard does not occur and social loss is zero.
- Moral hazard might not occur even in the presence of price distortion and behavioral changes if there is symmetric information between the customers and insurance companies.
  - However, symmetric information is not possible because insurance companies cannot monitor everything that patients do and price their actions accordingly.

- The information asymmetries that give rise to ex ante and ex post moral hazard are not identical.
  - In case of ex ante moral hazard (para gliding injury), some insurance companies do take some steps to monitor the behavior of their customers and provide incentive for less risky behavior.
  - Ex post moral hazard is driven by a slightly different information asymmetry concerning the necessity of any given medical treatment (miracle pill).
    - This information asymmetry, unlike the one discussed above, could actually be eliminated pretty easily in principle.
    - If the insurance company charged customers the full prices for their medical care, it would quickly determine which patients really needed expensive treatment and which patients were just taking advantage of artificially low prices.
    - The only problem with this strategy is it defeats the entire purpose of insurance, which is designed to reduce expenses in the case of illness or injury

- Moral hazard occurs if and only if three conditions hold:
    - The cost of a risky or wasteful action to an individual is reduced, usually as a consequence of insurance.
    - Asymmetric information prevents an insurer from adequately pricing the action.
    - That individual responds to the price distortion by changing his behavior—either by taking more risks or demanding more covered goods and services.

- How do health insurers limit moral hazard?
  - The extent of moral hazard depends on both how sensitive demand is to price and the amount of price distortion caused by insurance.
  - Insurers cannot alter customers' price sensitivity (which is a property of their demand functions), but they do have ways to reduce the price distortion due to insurance

Figure 6: Ways to limit moral hazard

- With full insurance, the marginal cost of health care to each consumer is zero, and each consumer will demand all the health care that provides him any positive utility at all.
- Even a procedure that costs the insurer millions but provides him only minimal utility, he demands.

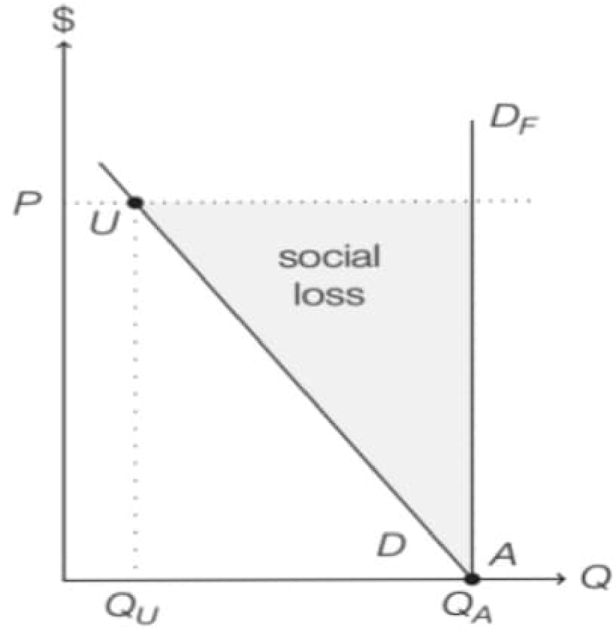

Figure 7: Full insurance

- Effective demand curve is D_F and he consumes at point Q_A
- Without insurance, the individual would consume at point Q_U.

- Coinsurance and copayment are two insurance contract provisions that maintain positive marginal costs for the insured.
  - Coinsurance: insurance provision in which enrollees pay a percentage of each medical bill, and the insurer covers the remaining portion.
  - Copayment: insurance provision in which enrollees pay only a fixed amount, called a copay
- These plans effectively limit insurance coverage so they are no longer full.

- Imagine the consumer starts at 0 percent coinsurance (full insurance from the previous slide).

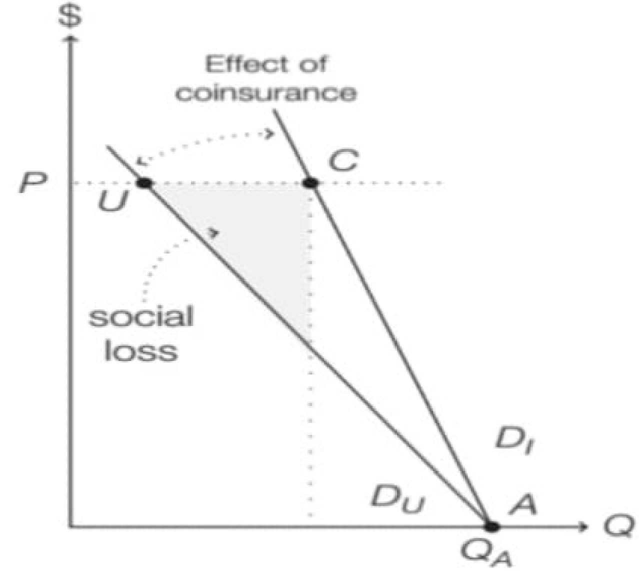

Figure 8: Coinsurance plan

- Her insurer then increases the level of coinsurance.
- As coinsurance rises, out-of-pocket prices move closer to actual prices, and the demand curve rotates back toward the uninsured demand curve D_U.
- At the extreme, coinsurance of 100 percent is equivalent to no insurance.

- Imagine his insurer institutes a copay of P_C , which becomes the effective price for each episode of care.

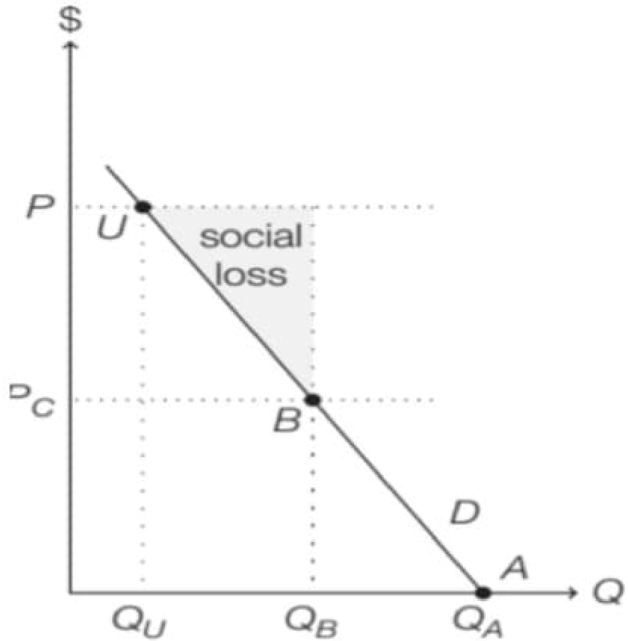

Figure 9: Copayment plan

- This reduces his demand from Q_A to Q_B.

- Imposition of either a copay or coinsurance increases the marginal cost of additional medical care to the consumer and lowers the quantity demanded.
- In both the cases, social loss reduces at the expense of increasing uncertainty faced by consumers who are no longer fully insured.

- In addition to coinsurance and copayment, many insurers also include deductibles as part of their offered plans.
- Deductibles set minimal levels of expenses below which the insurer does not help reimburse medical expenses.
  - Example: A person insured with a deductible of $ 1,000 pays for his first thousand dollars of health care expenditures out-of-pocket. His insurance policy then helps pay for expenses beyond the thousandth dollar.

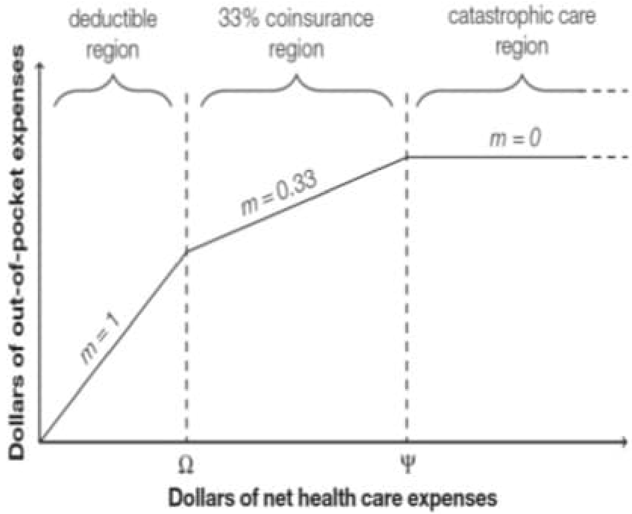

Figure 10: Relationship between out-of-pocket expenses and total medical expenditures

- This shows an insurance policy for a 33 percent coinsurance with a deductible of Ω and full insurance for catastrophic care above Ψ.

- Like coinsurance and copayments, deductibles detract from the fullness of insurance contracts and therefore limit the utility gained by consumers from insurance.
  - Lower deductibles mean fuller insurance but also more danger of moral hazard, so policies with lower deductibles tend to charge higher premiums.

- Some insurance companies try to observe and guide the preventative measures their customers take, while others choose to supervise the medical care that customers receive.
    - Motivation: Kaiser Permanente (KP), a major American insurer, has a program called Thrive that encourages enrollees to exercise and stay healthy. Its website offers online yoga courses, motivational tips for dieting, and a tool to find local farmer's markets.
- Employee incentive programs with payouts if you see a nutritionist, do yoga once a week or get a fitness test.
- Gatekeeping: a primary care physician to assess patients and decide how much care they really need.

- Evidence of moral hazard in health insurance:
- Ex ante moral hazard: It occurs when those with insurance coverage take more risks.
- RAND HIE: people on the free plan more likely to show up at the hospital with broken bones or drug abuse.

	Annual visits per 10,000 enrollees		
Diagnosis	Free plan	Cost-sharing plans	Ratio of free to cost-sharing
Fracture/dislocation	168	134	125%
Misc. serious trauma	67	57	118%
Acute alcohol/drug related	27	20	135%

Figure 11: Insurance policies with deductibles and copayments.

- It is also possible that this effect is entirely driven by ex post moral hazard – perhaps fully insured individuals are no more likely to suffer acute trauma, just more likely to show up at the hospital for treatment because it is cheaper for them, or more likely to show up with very minor versions of these injuries.

- In Ghana, malaria is the leading cause of morbidity and mortality, and accounts for about 38 percent of the country's outpatient visits and 36 percent of its admissions into health care facilities (Yilma et al. 2012).
- Yilma et al. (2012) find that insured households tend to have fewer members sleeping under treated mosquito nets than uninsured households.
- Case of preventative health care
  - If people randomly assigned to full insurance coverage are less likely to use preventative care, we could point to that as an instance of ex ante moral hazard – those people are taking more risks with their health because they know insurance has covered them.
  - If instead those people are using more preventative care, because the insurance company covers most of those costs as well. Then that would seem like an example of ex post moral hazard resulting from health care overconsumption.

- Low-income Mexicans whose villages were randomly included in the Seguro Popular program tended to seek less preventative care (flu shot or cancer screenings) than their compatriots who were randomly assigned to uninsurance.
- It is possible that uninsured low-income Mexicans were more invested in preventative care, probably because getting sick without insurance is worse than getting sick with insurance.
- By contrast, the RAND and Oregon studies find no evidence that insured people sought less preventative care like vaccinations or mammograms. Those with more insurance coverage tended to use more preventative care, probably because it was cheaper for them

- Studies are better able to observe ex post moral hazard than ex ante moral hazard in health insurance.
    - Ex ante moral hazard is usually reflected in personal behaviors that are hard to observe
    - Ex post moral hazard can be identified by comparing groups with different insurance coverage levels and studying detailed records of what treatments or benefits the participants receive.
- Stanford employees: after a 1967 change that required a new 25 percent copay led to a fall in the visits to the doctor by 24 percent (Scitovsky and Snyder 1972; Scitovsky and McCall 1977).
- RAND HIE: Higher rates of coinsurance were associated with fewer outpatient episodes and lower probability of an inpatient visit.
- Oregon Medicaid study: Lottery winners with the opportunity of enrolling in the state-sponsored insurance tended to use outpatient care more than lottery losers
    - If the increased use of health care due to lower prices improved health status, then the evidence found by both the RAND HIE and the OME might not be indications of moral hazard but rather of efficient use. However, this was not the case.

- Ortmann (2011), using data from Germany's small private health insurance industry, examines the impact of deductible size on expenditures for outpatient services.
- His model suggests that deductibles in Germany are associated with significant cost savings, up to 35 percent of total outpatient expenditures.
- Ortmann also finds that introducing a deductible can lead to reductions in health insurance premiums that actually exceed the amount of the deductible, leading to a decline in total payments for all customers.

- All Canadian residents have access to free appointments with a primary care physician, but only some Canadians have prescription drug coverage (often obtained through an employer or parent).
- Devlin et al. (2011) find that those who rarely visit the doctor start visiting the doctor much more often when they obtain prescription drug coverage.
- They speculate that this is because patients begin to substitute prescription drugs for over-the-counter drugs when they no longer have to pay their full price.
- One type of insurance created moral hazard for a different type of service.

- Eldridge et al. (2010) analyze data from the Australian National Health Survey and study those who opt to pay for private insurance that grants them access to more luxurious private hospitals.
- These patients receive perks like private rooms and their choice of doctor, so they might be expected to take advantage of these luxuries and extend their hospital stays.
- However, these patients did not seem to stay in the hospital any longer than otherwise similar patients who stayed in public hospitals.
- This is possible if patients feel that spending a night in the hospital is a fairly miserable experience even if it is free, then patients will seek to be discharged as soon as possible and will not react much to price changes.

- Moral hazard creates a kind of externality, because the costs of each person's risky decisions are borne by other people in the insurance pool.
- Given the social loss associated with moral hazard (by-product of insurance contracts), should insurance policies that create moral hazard be prohibited?
    - No, because insurance itself provides positive welfare gains that may offset the harm from moral hazard.
- Asymmetric information creates a tradeoff between more insurance coverage, which generates more moral hazard, and less insurance coverage, which increases risk exposure.